

Kernighan-Lin refinement

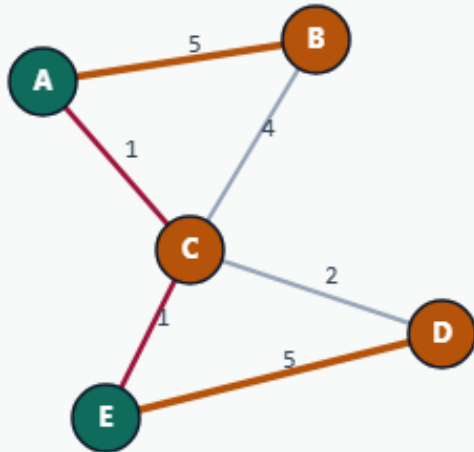
Kernighan-Lin improves an existing bipartition by swapping pairs across the cut

Bad seed: cutsize 12

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Bad seed: cutsize 12

Step 1 / 5 · bad-seed · module02-05-kernighan-lin



Explanation

Start from a terrible bipartition $AE|BCD$. Both heavy edges $A-B$ and $D-E$ are cut, so cutsize is 12. KL will search improving swaps.

- Seed parts: A, E vs B, C, D
- Cut includes $A-B(5)$ and $D-E(5)$
- Goal: reduce cut without enumerating all partitions

```
seed: {"A": "0", "E": "0", "B": "1", "C": "1", "D": "1"}  
cutsize: 12
```

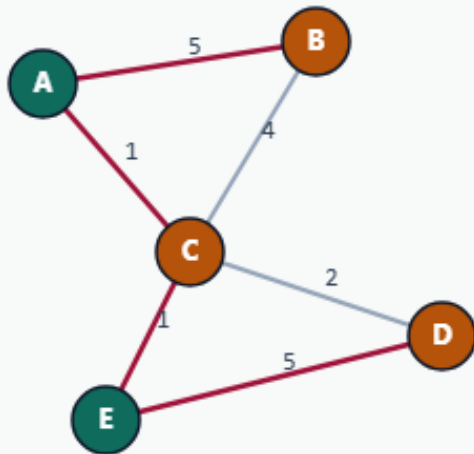
Bad seed: cutsize 12

Score pairwise swaps by gain

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Score pairwise swaps by gain

Step 2 / 5 · gain-idea · module02-05-kernighan-lin



Explanation

KL considers swapping one vertex from each side. Gain estimates how much the cut shrinks. The best unlocked pair here is $A \leftrightarrow D$ with gain 9.

- $D(v) = \text{external} - \text{internal}$ for each vertex
- Swap gain uses D values and the edge between the pair
- Lock pairs after considering them in a pass

Best candidate swap: $A \leftrightarrow D$
gain $g = 9$

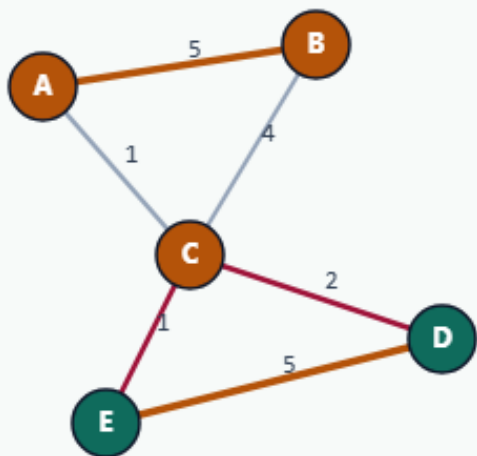
Score pairwise swaps by gain

Accept prefix: only $A \leftrightarrow D$

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Accept prefix: only $A \leftrightarrow D$

Step 3 / 5 · accept-swap · module02-05-kernighan-lin



Explanation

Pass 0 builds a sequence of candidate swaps, then keeps the prefix with best cumulative gain. Here $\text{best_k}=1$: perform $A \leftrightarrow D$ once.

- $\text{best_k} = 1$
- $\text{bestCum} = 9$
- cut $12 \rightarrow 3$ in one swap

```
pass 0: A/D(9)
cutBefore=12 cutAfter=3
improved=true
```

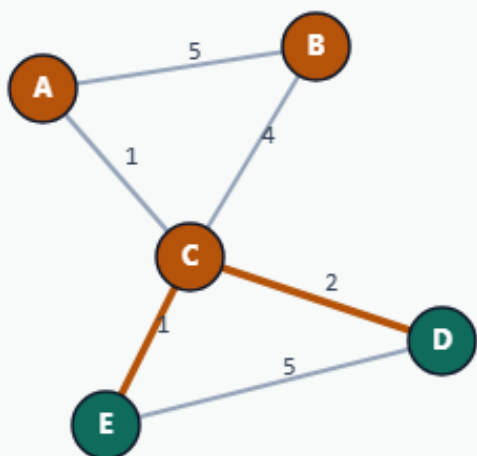
Accept prefix: only $A \leftrightarrow D$

Refined partition ABC|DE

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Refined partition ABC|DE

Step 4 / 5 · final · module02-05-kernighan-lin



Explanation

After the swap, A joins B,C and D joins E. Heavy edges are internal; only the weak C-D/C-E bridge remains cut.

- Final parts: ABC|DE
- Matches the greedy/LP communities
- Cutsizes golden: 3

```
cutsizes: 3  
parts: ABC|DE
```

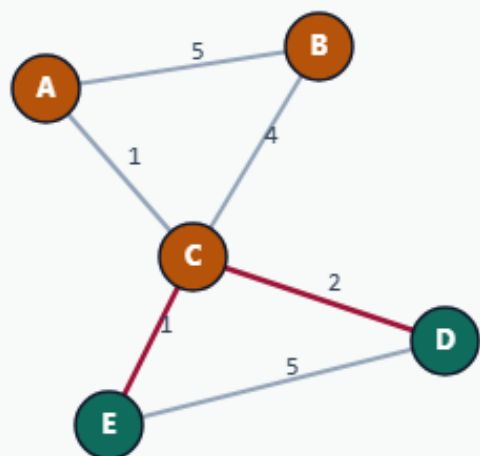
Refined partition ABC|DE

Next pass finds nothing

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Next pass finds nothing

Step 5 / 5 · pass1-stop · module02-05-kernighan-lin



Explanation

Pass 1 reports improved=false. KL stops when a full pass cannot improve—local optimum for swap moves from this seed.

- Local, not global, optimum
- Quality depends on the seed
- Still the classic bipartition refiner

```
pass 1: best_k=0 improved=false  
stop
```

Next pass finds nothing

Browser lab track

- In the browser lab, show the seed only, then run KL
- Check the challenges for cutsize twelve, the A-D swap, and the final cutsize three

Implement track

- Load the tiny graph and the bad seed
- Print cutsize twelve before you start
- Run the reference KL solver and confirm it accepts A-D

Implement track — try these

```
# pwd — print working directory  
pwd
```

```
# ls examples — graph + seed partition  
ls examples
```

```
# run full KL refinement on the bad seed  
export PYTHONPATH=../common  
python ../common/solvers.py examples/tiny_graph.json --mode kl  
--seed examples/seed_partition.json
```

Pitfalls to watch

- Forgetting rollback is the classic bug
- Stale D-values after a swap pick the wrong pair next
- And if your seed isn't labeled zero and one, the gain math won't match

Your turn

- Reproduce cut twelve to three with the accepted A-D swap
- Finish the checklist and quiz